PRACTICAL NO. 4

Advanced Sales Analytics and Visualization

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Course: Data Analytics and Visualization

Course Code: CT-471

Practical Overview

This advanced practical covers Power BI's AI-enabled and distribution visuals for strategic analytics. Distribution visuals (Pie, Donut, Treemap) reveal market concentration and structural patterns. Slicers enable dynamic filtering. Decomposition Trees and Key Influencers leverage AI to identify drivers. Q&A visuals support natural language exploration. KPI and Gauge measure performance momentum.

Visual Category	Questions
Pie Chart (Distribution)	Q1, Q2
Donut Chart (Segmentation)	Q3, Q4
Treemap (Hierarchical Concentration)	Q5, Q6, Q7, Q8
Slicer (Dynamic Filtering)	Q9, Q10
Decomposition Tree (AI Drill-down)	Q11, Q12, Q13
Key Influencers (AI Driver Analysis)	Q14, Q15, Q16
Q&A Visual (Natural Language)	Q17, Q18
KPI & Gauge (Performance)	Q19, Q20

Questions 1–20

Q1 – Pie Chart (Pareto)

Task: Pareto Category Contribution

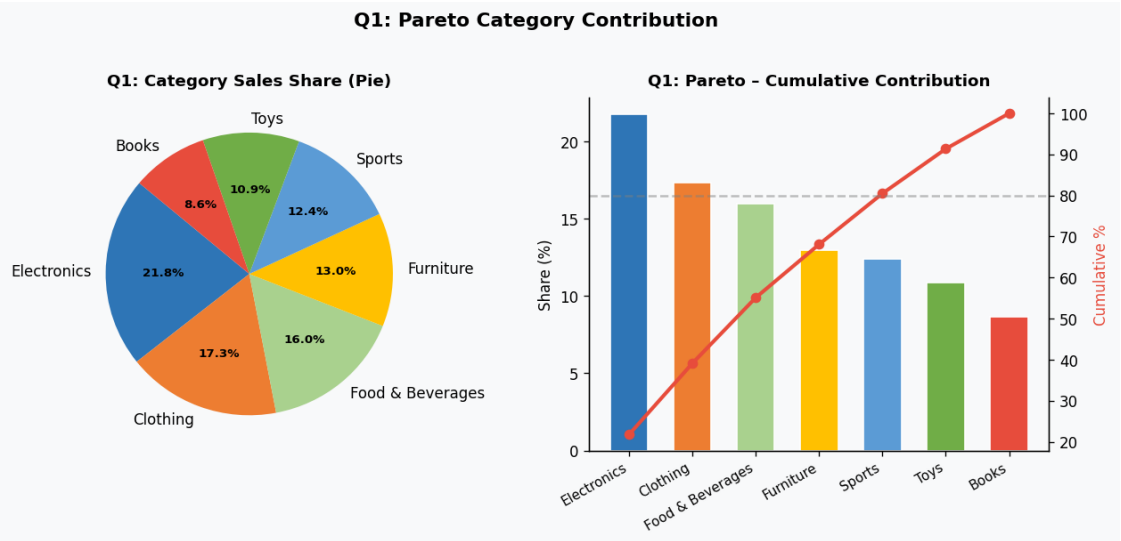
Pie chart showing each category's share of total sales, paired with a Pareto curve showing cumulative percentage to identify which categories contribute 80% of revenue.

SQL Query

```
WITH CategorySales AS (
  SELECT C.CATEGORY_NAME,
         SUM(F.SALES_AMOUNT)                                    AS TOTAL_SALES
  FROM   SALES_FACT F
  JOIN   CATEGORY_DIM C ON F.CAT_KEY = C.CAT_KEY
  GROUP  BY C.CATEGORY_NAME
),
Ranked AS (
  SELECT *,
         ROUND(TOTAL_SALES * 100.0 / SUM(TOTAL_SALES) OVER (), 2)   AS SALES_SHARE_PCT,
         ROUND(SUM(TOTAL_SALES) OVER (ORDER BY TOTAL_SALES DESC
               ROWS UNBOUNDED PRECEDING) * 100.0 /
               SUM(TOTAL_SALES) OVER (), 2)                          AS CUMULATIVE_PCT
  FROM   CategorySales
)
```

```
SELECT CATEGORY_NAME, TOTAL_SALES, SALES_SHARE_PCT, CUMULATIVE_PCT
FROM   Ranked
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Pie Chart (Pareto)



Q2 – Pie Chart (Regional Concentration)

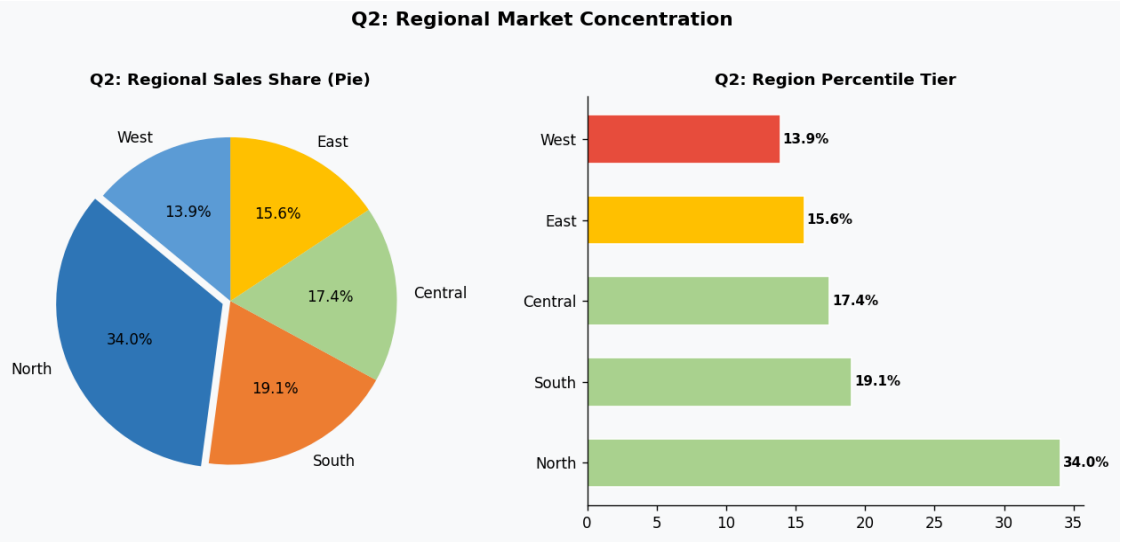
Task: Regional Market Concentration

Pie chart showing regional market concentration, with percentile ranking to classify regions as dominant (top quartile) vs emerging.

SQL Query

```
WITH RegionSales AS (
  SELECT G.REGION_NAME,
         SUM(F.SALES_AMOUNT)                                        AS TOTAL_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY = G.GEO_KEY
  GROUP  BY G.REGION_NAME
),
Ranked AS (
  SELECT *,
         ROUND(TOTAL_SALES * 100.0 / SUM(TOTAL_SALES) OVER (), 2)    AS SALES_SHARE_PCT,
         PERCENT_RANK() OVER (ORDER BY TOTAL_SALES)                   AS PERCENTILE_RANK
  FROM   RegionSales
)
SELECT REGION_NAME, TOTAL_SALES, SALES_SHARE_PCT,
       ROUND(PERCENTILE_RANK * 100, 1)  AS PERCENTILE_RANK
FROM   Ranked
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Pie Chart (Regional Concentration)



Q3 – Donut Chart (Seasonal Volatility)

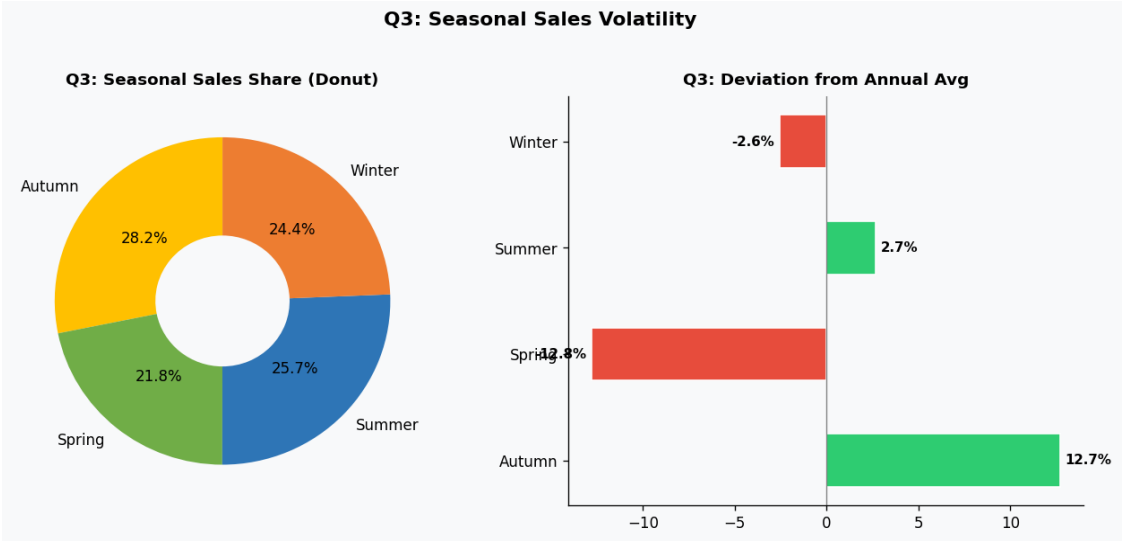
Task: Seasonal Volatility Share

Donut chart visualizing seasonal sales shares alongside a deviation bar chart showing how each season compares to the annual average.

SQL Query

```
WITH SeasonalSales AS (
  SELECT CASE
           WHEN T.MONTH_NUMBER IN (12,1,2) THEN 'Winter'
           WHEN T.MONTH_NUMBER IN (3,4,5)  THEN 'Spring'
           WHEN T.MONTH_NUMBER IN (6,7,8)  THEN 'Summer'
           ELSE 'Autumn'
         END                                                    AS SEASON,
         SUM(F.SALES_AMOUNT)                                    AS SEASONAL_SALES
  FROM   SALES_FACT F
  JOIN   TIME_DIM T ON F.TIME_KEY = T.TIME_KEY
  GROUP  BY SEASON
)
SELECT SEASON, SEASONAL_SALES,
       ROUND(SEASONAL_SALES * 100.0 / SUM(SEASONAL_SALES) OVER (), 2)  AS
SEASON_SHARE_PCT,
       ROUND(SEASONAL_SALES - AVG(SEASONAL_SALES) OVER (), 2)          AS DEV_FROM_AVG
FROM   SeasonalSales
ORDER  BY SEASONAL_SALES DESC;
```

Power BI Visual: Donut Chart (Seasonal Volatility)



Q4 – Donut Chart (Long Tail)

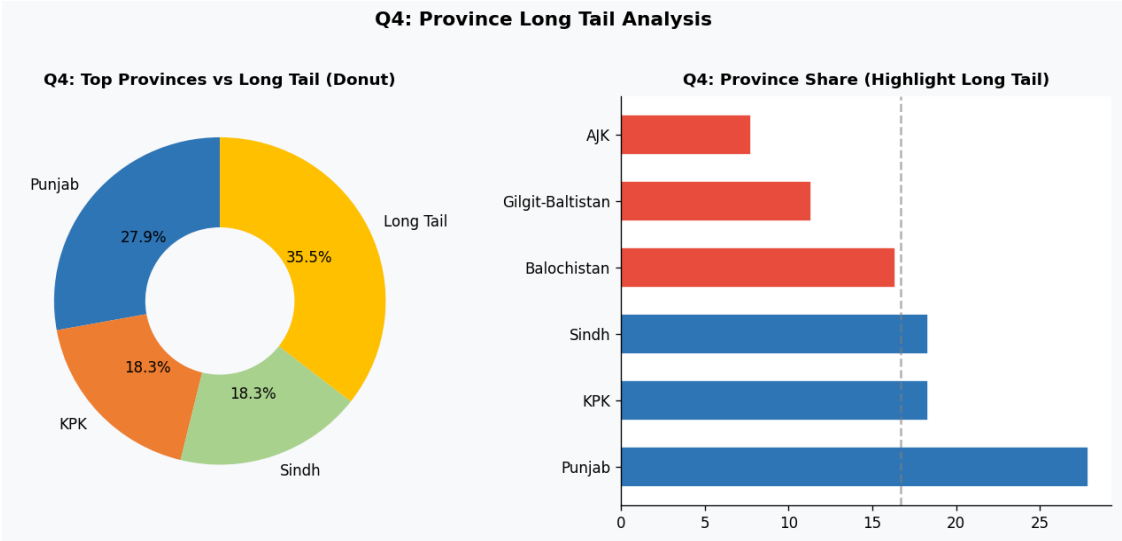
Task: Top Provinces vs Long Tail Analysis

Donut chart grouping top provinces individually and consolidating lower-ranked provinces into a 'Long Tail' segment to reveal the 80/20 distribution.

SQL Query

```
WITH ProvinceSales AS (
  SELECT G.STATE_PROVINCE_NAME,
         SUM(F.SALES_AMOUNT)                                          AS TOTAL_SALES,
         ROUND(SUM(F.SALES_AMOUNT)*100.0/SUM(SUM(F.SALES_AMOUNT)) OVER(),2) AS
SALES_SHARE
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  GROUP  BY G.STATE_PROVINCE_NAME
),
Classified AS (
  SELECT *,
         PERCENT_RANK() OVER (ORDER BY TOTAL_SALES DESC)              AS PCTILE
  FROM   ProvinceSales
)
SELECT STATE_PROVINCE_NAME, TOTAL_SALES, SALES_SHARE,
       CASE WHEN PCTILE <= 0.25 THEN STATE_PROVINCE_NAME
            ELSE 'Long Tail' END                                      AS DISPLAY_NAME
FROM   Classified
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Donut Chart (Long Tail)



Q5 – Treemap (HHI Concentration)

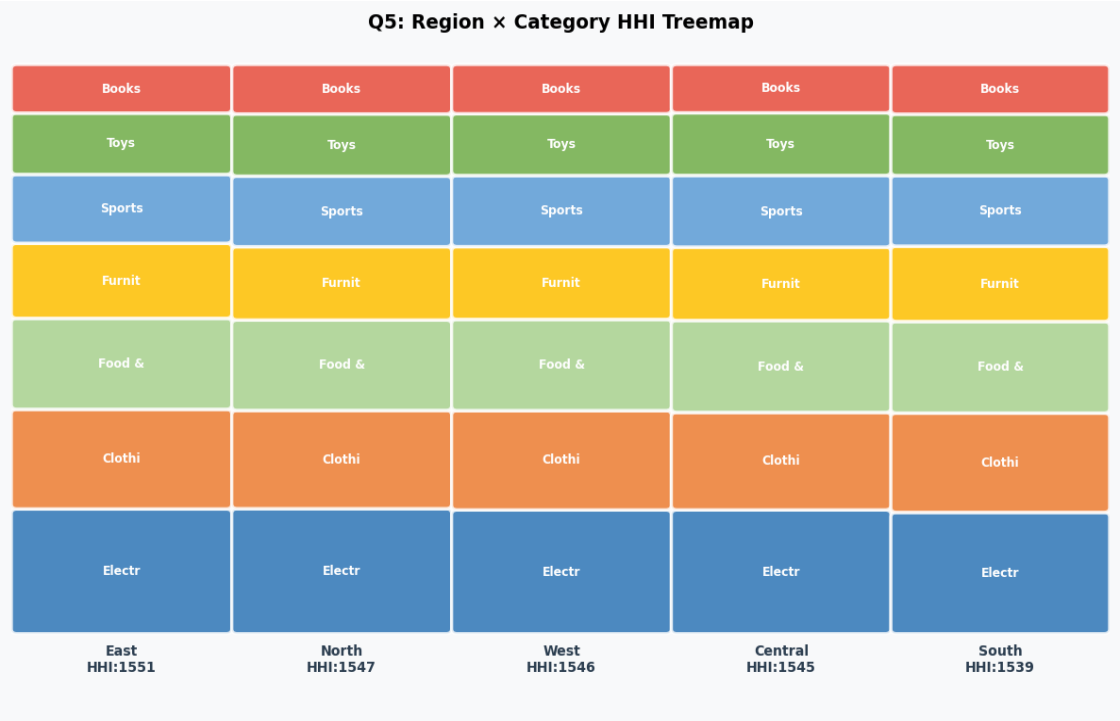
Task: Region x Category Concentration Index (HHI)

Treemap visualizing region-category sales hierarchy with each region's Herfindahl-Hirschman Index (HHI) showing market concentration level.

SQL Query

```
WITH RegionCatSales AS (
  SELECT G.REGION_NAME, C.CATEGORY_NAME,
         SUM(F.SALES_AMOUNT)                                        AS TOTAL_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  GROUP  BY G.REGION_NAME, C.CATEGORY_NAME
),
ShareCalc AS (
  SELECT *,
         ROUND(TOTAL_SALES*100.0/SUM(TOTAL_SALES) OVER (PARTITION BY REGION_NAME),2) AS
CAT_SHARE_PCT
  FROM   RegionCatSales
),
HHI AS (
  SELECT REGION_NAME,
         ROUND(SUM(POWER(CAT_SHARE_PCT,2)),0)                       AS HHI_INDEX
  FROM   ShareCalc
  GROUP  BY REGION_NAME
)
SELECT s.REGION_NAME, s.CATEGORY_NAME, s.TOTAL_SALES, s.CAT_SHARE_PCT, h.HHI_INDEX
FROM   ShareCalc s JOIN HHI h ON s.REGION_NAME=h.REGION_NAME
ORDER  BY h.HHI_INDEX DESC, s.TOTAL_SALES DESC;
```

Power BI Visual: Treemap (HHI Concentration)



Q6 – Treemap (YoY Growth)

Task: Multi-Level Growth Comparison

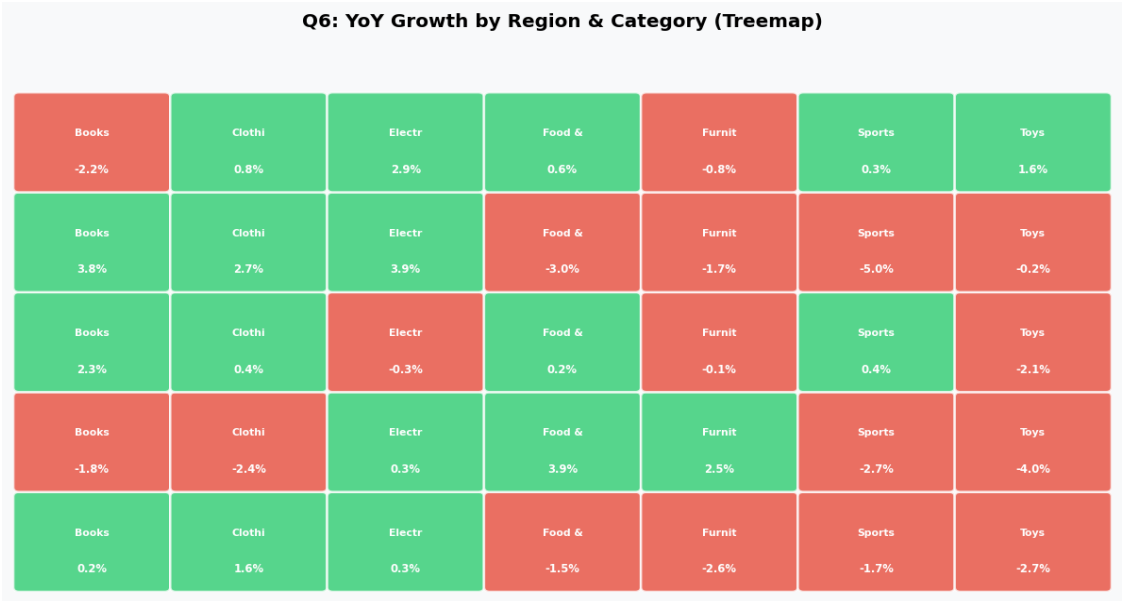
Treemap representing year-over-year growth at region-category level — green tiles for positive growth, red for decline.

SQL Query

```
WITH YoYData AS (
  SELECT G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME,
         SUM(F.SALES_AMOUNT) AS YR_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME
)
SELECT REGION_NAME, CATEGORY_NAME,
       MAX(CASE WHEN YEAR_NAME=2023 THEN YR_SALES END)              AS SALES_2023,
       MAX(CASE WHEN YEAR_NAME=2024 THEN YR_SALES END)              AS SALES_2024,
       ROUND((MAX(CASE WHEN YEAR_NAME=2024 THEN YR_SALES END) -
              MAX(CASE WHEN YEAR_NAME=2023 THEN YR_SALES END))*100.0 /
             NULLIF(MAX(CASE WHEN YEAR_NAME=2023 THEN YR_SALES END),0),2) AS
YOY_GROWTH_PCT
```

```
FROM   YoYData
GROUP  BY REGION_NAME, CATEGORY_NAME
ORDER  BY REGION_NAME, YOY_GROWTH_PCT DESC;
```

Power BI Visual: Treemap (YoY Growth)



Q7 – Treemap (Volume vs Revenue)

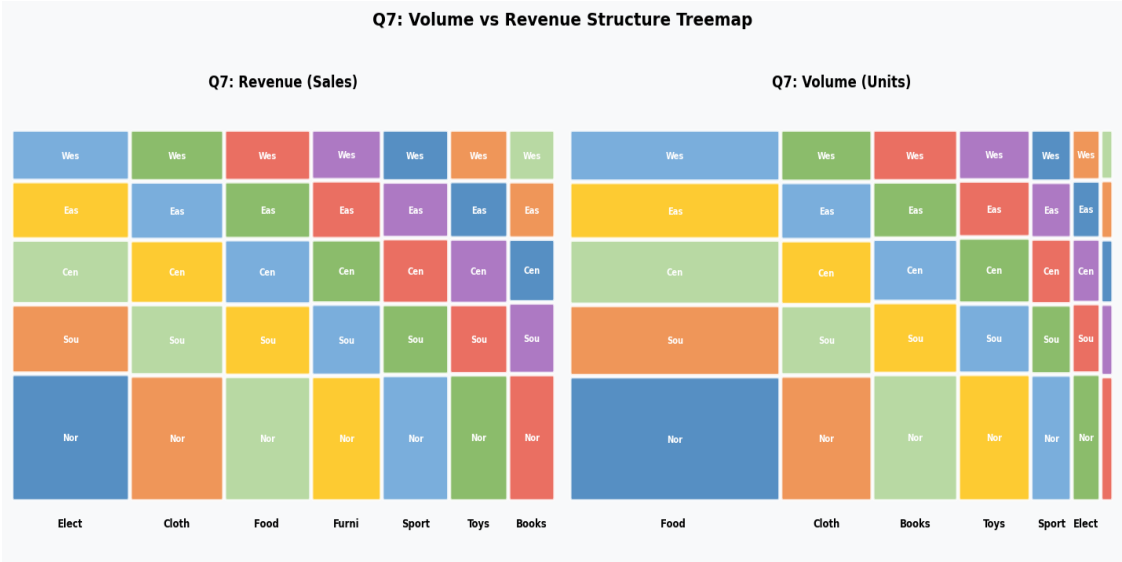
Task: Volume vs Revenue Structure

Side-by-side treemaps comparing revenue share vs volume share across region-category combinations to reveal revenue-volume misalignments.

SQL Query

```
SELECT G.REGION_NAME, C.CATEGORY_NAME,
       SUM(F.SALES_AMOUNT)                                          AS TOTAL_SALES,
       SUM(F.UNITS_SOLD)                                           AS TOTAL_UNITS,
       ROUND(SUM(F.SALES_AMOUNT)*100.0/SUM(SUM(F.SALES_AMOUNT)) OVER(),2) AS
REVENUE_SHARE_PCT,
       ROUND(SUM(F.UNITS_SOLD)*100.0/SUM(SUM(F.UNITS_SOLD)) OVER(),2)    AS
VOLUME_SHARE_PCT
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
GROUP  BY G.REGION_NAME, C.CATEGORY_NAME
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Treemap (Volume vs Revenue)



Q8 – Treemap (Rolling 3-Month)

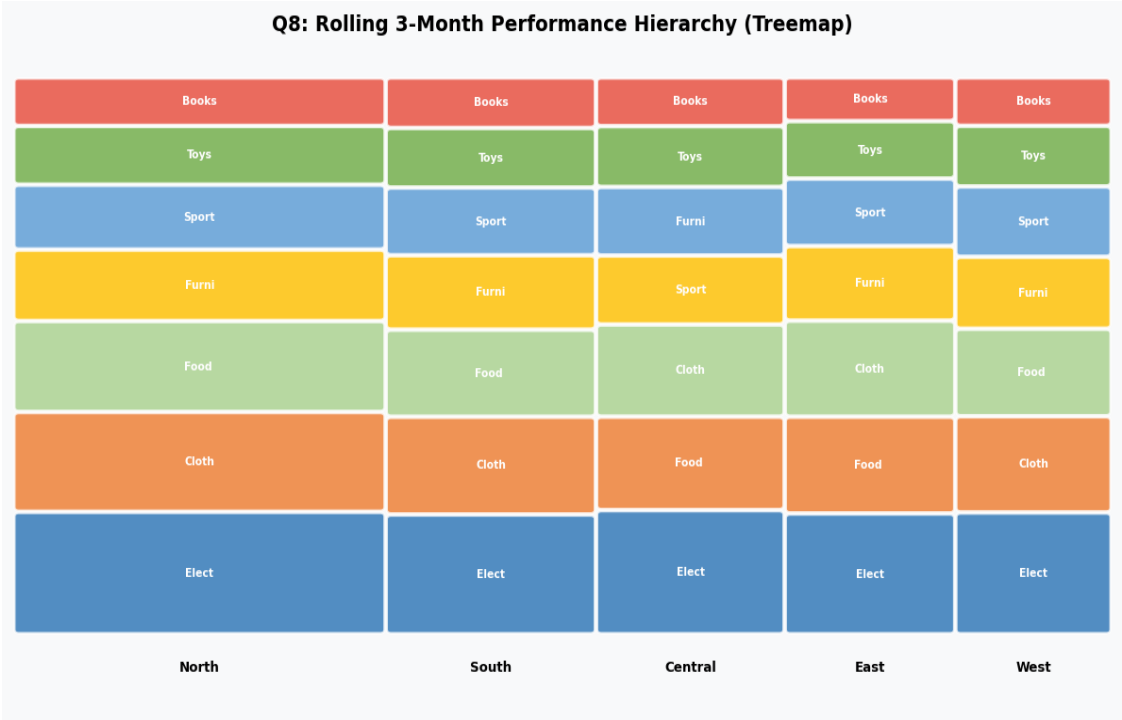
Task: Rolling 3-Month Performance Hierarchy

Treemap showing rolling 3-month sales hierarchy at region-category level, using only the three most recent months for currency.

SQL Query

```
SELECT G.REGION_NAME, C.CATEGORY_NAME,
       T.YEAR_NAME, T.MONTH_NUMBER,
       SUM(F.SALES_AMOUNT)                                          AS MONTHLY_SALES,
       SUM(SUM(F.SALES_AMOUNT))
         OVER (PARTITION BY G.REGION_NAME, C.CATEGORY_NAME
               ORDER BY T.YEAR_NAME, T.MONTH_NUMBER
               ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)              AS ROLLING_3M_SALES
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER
ORDER  BY T.YEAR_NAME, T.MONTH_NUMBER;
```

Power BI Visual: Treemap (Rolling 3-Month)



Q9 – Slicer (Performance Tiering)

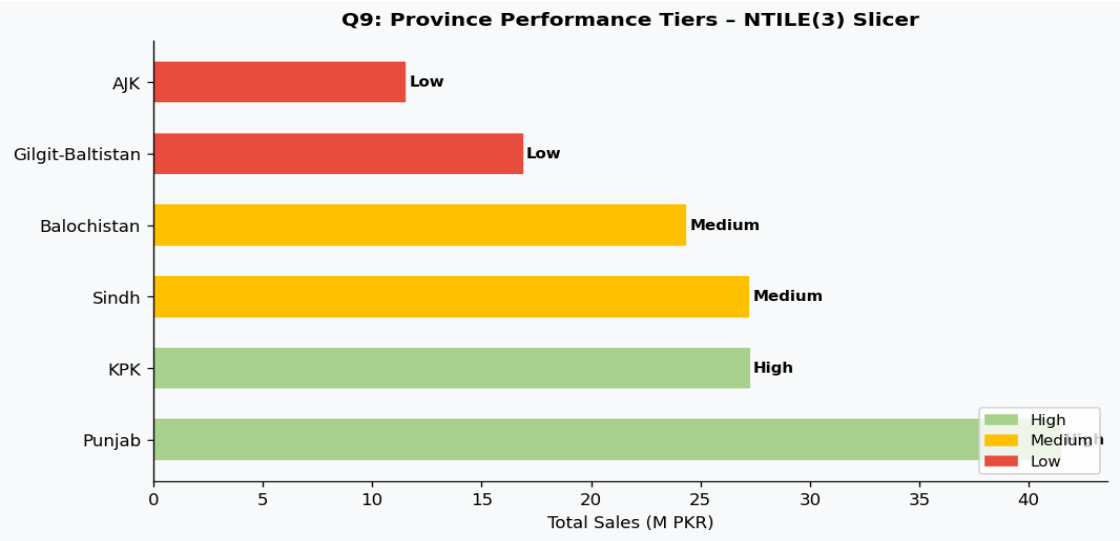
Task: Dynamic Performance Tiering

Bar chart classifying provinces into High/Medium/Low tiers using NTILE(3) percentile logic — enables dynamic slicer filtering by tier.

SQL Query

```
WITH ProvinceMetrics AS (
  SELECT G.STATE_PROVINCE_NAME,
         SUM(F.SALES_AMOUNT)                                    AS TOTAL_SALES,
         NTILE(3) OVER (ORDER BY SUM(F.SALES_AMOUNT))          AS PERFORMANCE_TILE
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  GROUP  BY G.STATE_PROVINCE_NAME
)
SELECT STATE_PROVINCE_NAME, TOTAL_SALES,
       CASE PERFORMANCE_TILE
         WHEN 3 THEN 'High'
         WHEN 2 THEN 'Medium'
         ELSE 'Low'
       END                                                      AS SALES_TIER
FROM   ProvinceMetrics
ORDER  BY TOTAL_SALES DESC;
```

Power BI Visual: Slicer (Performance Tiering)



Q10 – Slicer (Volatility Filtering)

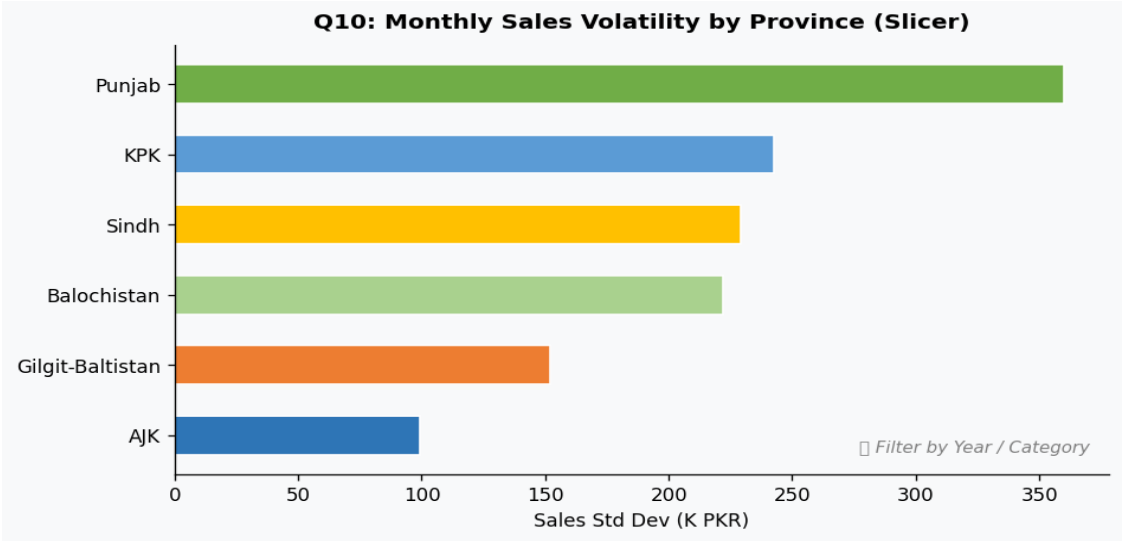
Task: Time-Based Volatility Filtering

Slicer dataset with province-level monthly sales volatility (standard deviation) filterable by year and category.

SQL Query

```
WITH MonthlyProvince AS (
  SELECT G.STATE_PROVINCE_NAME, T.YEAR_NAME, T.MONTH_NUMBER,
         C.CATEGORY_NAME,
         SUM(F.SALES_AMOUNT)                                        AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  GROUP  BY G.STATE_PROVINCE_NAME, T.YEAR_NAME, T.MONTH_NUMBER, C.CATEGORY_NAME
)
SELECT STATE_PROVINCE_NAME, YEAR_NAME, CATEGORY_NAME,
       ROUND(STDDEV(MONTHLY_SALES) OVER
             (PARTITION BY STATE_PROVINCE_NAME, CATEGORY_NAME), 2)  AS SALES_STD_DEV,
       ROUND(AVG(MONTHLY_SALES) OVER
             (PARTITION BY STATE_PROVINCE_NAME, CATEGORY_NAME), 2)  AS AVG_MONTHLY_SALES
FROM   MonthlyProvince;
```

Power BI Visual: Slicer (Volatility Filtering)



Q11 – Decomposition Tree (Growth Drivers)

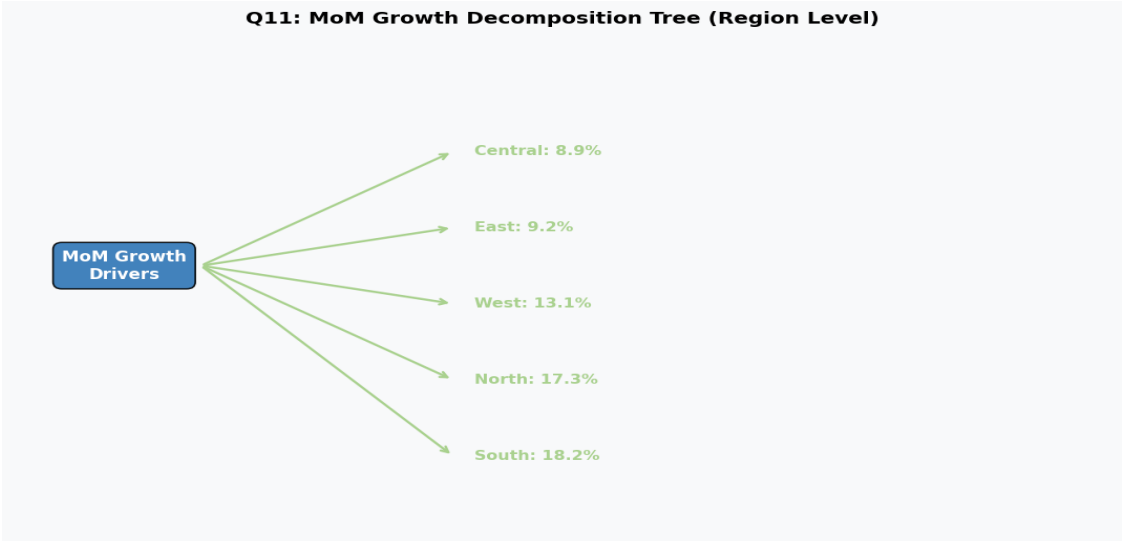
Task: Growth Driver Analysis

Decomposition tree dataset showing month-over-month growth with hierarchy drilldown across Region, Province, and Category.

SQL Query

```
WITH MonthlyGrowth AS (
  SELECT G.REGION_NAME, G.STATE_PROVINCE_NAME, C.CATEGORY_NAME,
         T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.SALES_AMOUNT)                                        AS MONTHLY_SALES,
         LAG(SUM(F.SALES_AMOUNT))
           OVER (PARTITION BY G.REGION_NAME, G.STATE_PROVINCE_NAME, C.CATEGORY_NAME
                 ORDER BY T.YEAR_NAME, T.MONTH_NUMBER)              AS PREV_MONTH_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.REGION_NAME, G.STATE_PROVINCE_NAME, C.CATEGORY_NAME,
            T.YEAR_NAME, T.MONTH_NUMBER
)
SELECT REGION_NAME, STATE_PROVINCE_NAME, CATEGORY_NAME,
       YEAR_NAME, MONTH_NUMBER, MONTHLY_SALES, PREV_MONTH_SALES,
       ROUND((MONTHLY_SALES - PREV_MONTH_SALES)*100.0 /
             NULLIF(PREV_MONTH_SALES,0), 2)                         AS MOM_GROWTH_PCT
FROM   MonthlyGrowth
WHERE  PREV_MONTH_SALES IS NOT NULL
ORDER  BY MOM_GROWTH_PCT DESC;
```

Power BI Visual: Decomposition Tree (Growth Drivers)



Q12 – Decomposition Tree (Revenue/Unit)

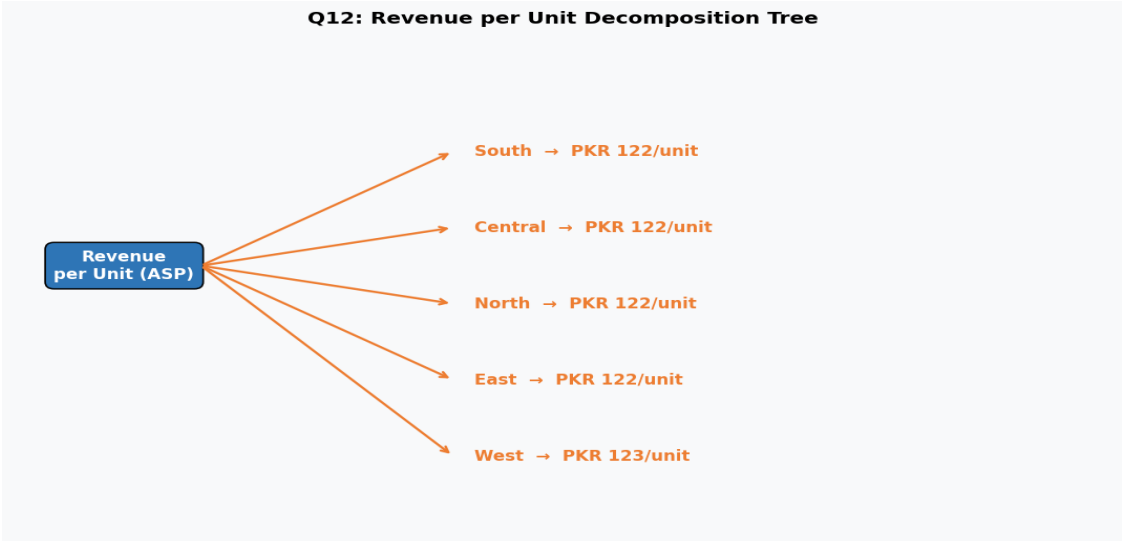
Task: Revenue per Unit Breakdown

Decomposition tree dataset for Average Selling Price breakdown, enabling hierarchical exploration from region down to category.

SQL Query

```
SELECT G.REGION_NAME, G.STATE_PROVINCE_NAME, C.CATEGORY_NAME,
       SUM(F.SALES_AMOUNT)                                          AS TOTAL_SALES,
       SUM(F.UNITS_SOLD)                                           AS TOTAL_UNITS,
       ROUND(SUM(F.SALES_AMOUNT)/NULLIF(SUM(F.UNITS_SOLD),0), 2)   AS AVG_SELLING_PRICE
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
GROUP  BY G.REGION_NAME, G.STATE_PROVINCE_NAME, C.CATEGORY_NAME
ORDER  BY AVG_SELLING_PRICE DESC;
```

Power BI Visual: Decomposition Tree (Revenue/Unit)



Q13 – Decomposition Tree (Decline Causes)

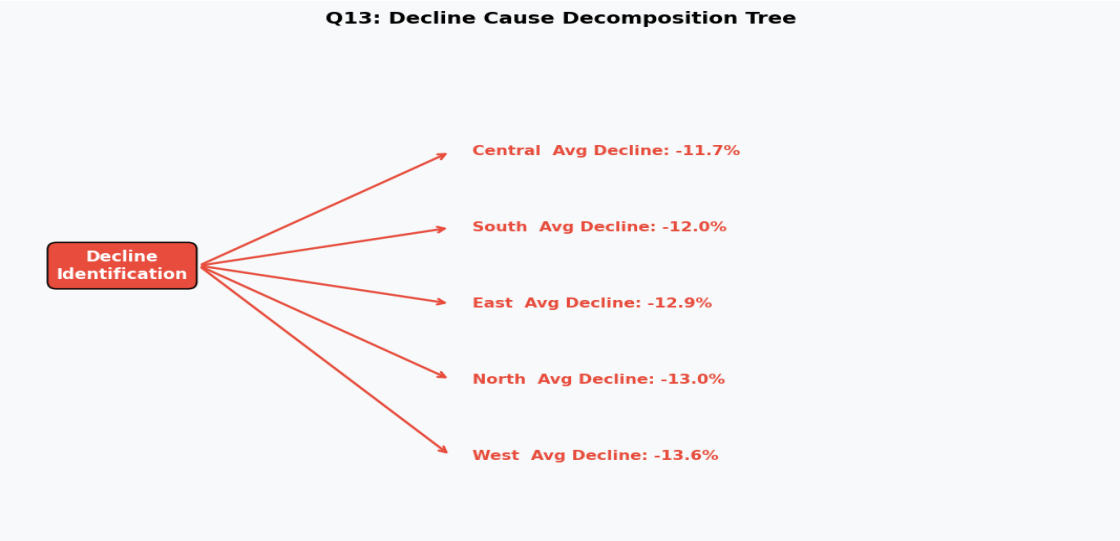
Task: Decline Cause Identification

Decomposition tree dataset isolating months with negative growth to identify which region-category combinations drive declines.

SQL Query

```
WITH MonthlyData AS (
  SELECT G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.SALES_AMOUNT)                                        AS MONTHLY_SALES,
         LAG(SUM(F.SALES_AMOUNT))
           OVER (PARTITION BY G.REGION_NAME, C.CATEGORY_NAME
                 ORDER BY T.YEAR_NAME, T.MONTH_NUMBER)              AS PREV_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER
)
SELECT REGION_NAME, CATEGORY_NAME, YEAR_NAME, MONTH_NUMBER,
       MONTHLY_SALES, PREV_SALES,
       ROUND((MONTHLY_SALES - PREV_SALES)*100.0/NULLIF(PREV_SALES,0),2) AS MOM_GROWTH
FROM   MonthlyData
WHERE  MONTHLY_SALES < PREV_SALES
ORDER  BY MOM_GROWTH ASC;
```

Power BI Visual: Decomposition Tree (Decline Causes)



Q14 – Key Influencers (High Sales)

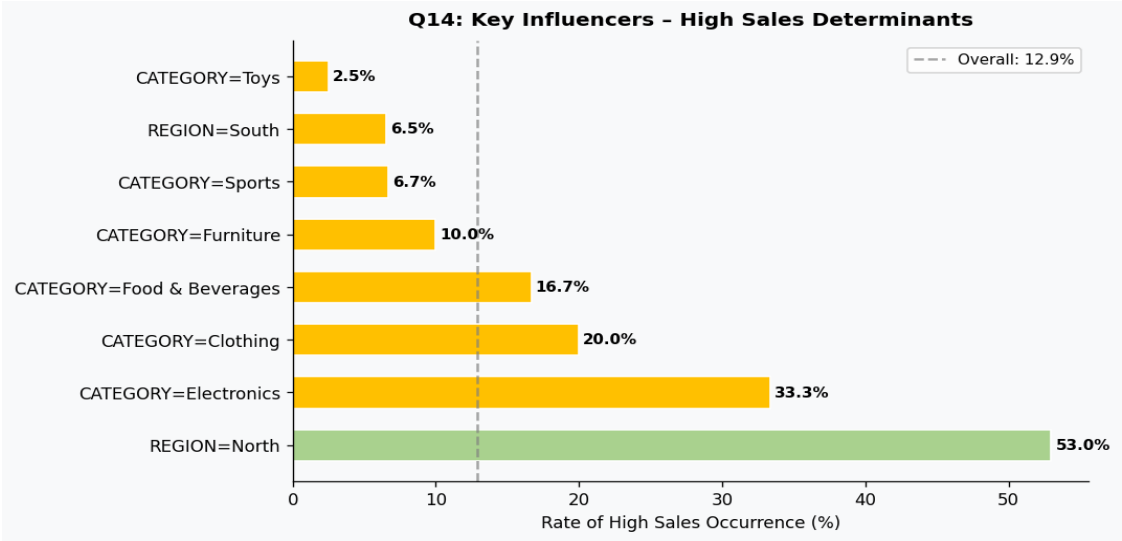
Task: High Sales Determinants

Key Influencers dataset classifying months as High Sales (above mean + 1 std dev) or Normal, with region and category factors.

SQL Query

```
WITH MonthlyStats AS (
  SELECT G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.SALES_AMOUNT)                                        AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER
),
Thresholds AS (
  SELECT AVG(MONTHLY_SALES) + STDDEV(MONTHLY_SALES)                 AS HIGH_THRESHOLD
  FROM   MonthlyStats
)
SELECT m.REGION_NAME, m.CATEGORY_NAME, m.YEAR_NAME, m.MONTH_NUMBER,
       m.MONTHLY_SALES,
       CASE WHEN m.MONTHLY_SALES >= t.HIGH_THRESHOLD
            THEN 'High Sales' ELSE 'Normal' END                     AS SALES_CLASS
FROM   MonthlyStats m CROSS JOIN Thresholds t
ORDER  BY m.MONTHLY_SALES DESC;
```

Power BI Visual: Key Influencers (High Sales)



Q15 – Key Influencers (Volume Surge)

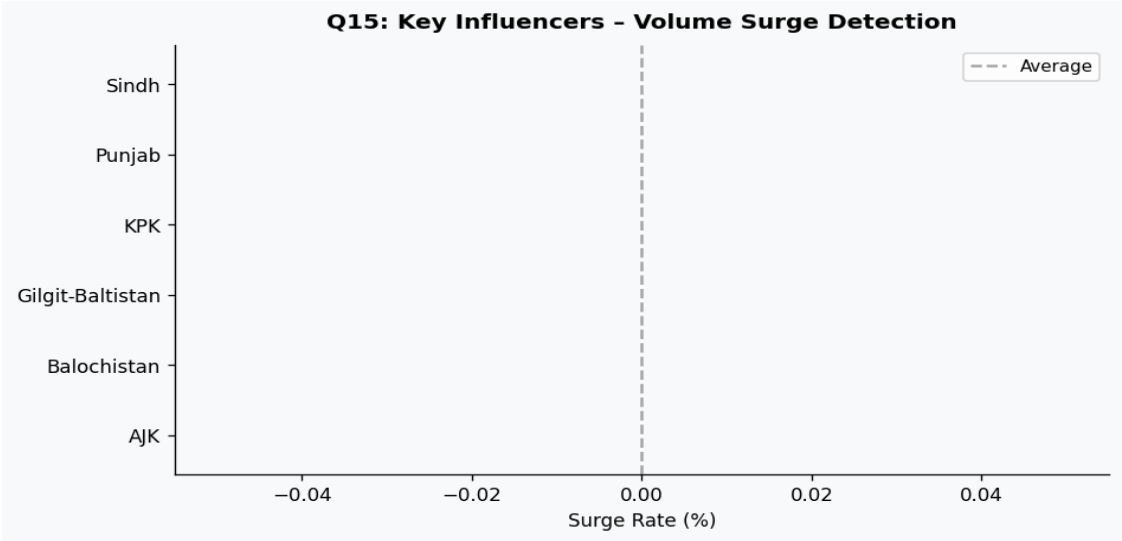
Task: Volume Surge Detection

Key Influencers dataset flagging provinces with volume surges (units > 120% of 3-month rolling average) for driver analysis.

SQL Query

```
WITH RollingUnits AS (
  SELECT G.STATE_PROVINCE_NAME, T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.UNITS_SOLD)                                          AS MONTHLY_UNITS,
         AVG(SUM(F.UNITS_SOLD))
           OVER (PARTITION BY G.STATE_PROVINCE_NAME
                 ORDER BY T.YEAR_NAME, T.MONTH_NUMBER
                 ROWS BETWEEN 2 PRECEDING AND CURRENT ROW)          AS ROLLING_3M_AVG
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.STATE_PROVINCE_NAME, T.YEAR_NAME, T.MONTH_NUMBER
)
SELECT STATE_PROVINCE_NAME, YEAR_NAME, MONTH_NUMBER,
       MONTHLY_UNITS, ROLLING_3M_AVG,
       CASE WHEN MONTHLY_UNITS > ROLLING_3M_AVG * 1.2
            THEN 'Volume Surge' ELSE 'Normal' END                   AS SURGE_FLAG
FROM   RollingUnits
ORDER  BY STATE_PROVINCE_NAME, YEAR_NAME, MONTH_NUMBER;
```

Power BI Visual: Key Influencers (Volume Surge)



Q16 – Key Influencers (Seasonal Impact)

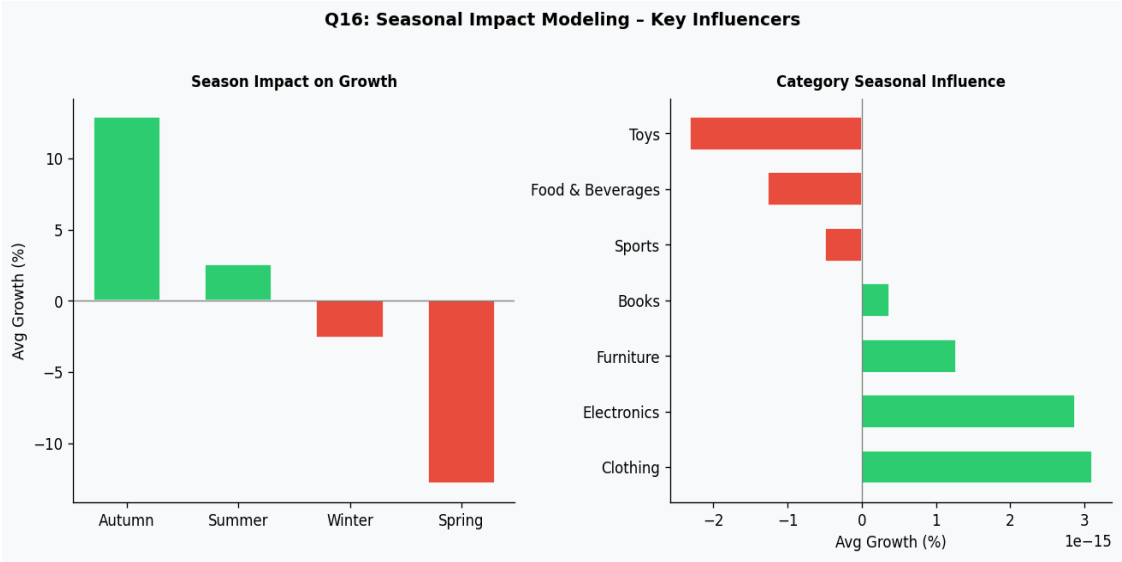
Task: Seasonal Impact Modeling

Key Influencers dataset combining Season, Region, Category, and seasonal growth rate to identify the strongest performance drivers.

SQL Query

```
SELECT G.REGION_NAME, C.CATEGORY_NAME,
       CASE WHEN T.MONTH_NUMBER IN (12,1,2) THEN 'Winter'
            WHEN T.MONTH_NUMBER IN (3,4,5)  THEN 'Spring'
            WHEN T.MONTH_NUMBER IN (6,7,8)  THEN 'Summer'
            ELSE 'Autumn' END                                    AS SEASON,
       SUM(F.SALES_AMOUNT)                                      AS SEASONAL_SALES,
       ROUND((SUM(F.SALES_AMOUNT) - AVG(SUM(F.SALES_AMOUNT))
              OVER (PARTITION BY G.REGION_NAME, C.CATEGORY_NAME))*100.0 /
             NULLIF(AVG(SUM(F.SALES_AMOUNT))
              OVER (PARTITION BY G.REGION_NAME, C.CATEGORY_NAME),0),2) AS
SEASONAL_GROWTH_PCT
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, SEASON;
```

Power BI Visual: Key Influencers (Seasonal Impact)



Q17 – Q&A Visual (Growth & Share Dataset)

Task: Growth and Share Dataset Preparation

Semantic Q&A dataset with Sales, Market Share %, YoY Growth %, and ASP per region-category-year combination.

SQL Query

```
SELECT G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME,
       SUM(F.SALES_AMOUNT)                                          AS TOTAL_SALES,
       ROUND(SUM(F.SALES_AMOUNT)*100.0/SUM(SUM(F.SALES_AMOUNT)) OVER(PARTITION BY
T.YEAR_NAME),2) AS MARKET_SHARE_PCT,
       ROUND((SUM(F.SALES_AMOUNT) - LAG(SUM(F.SALES_AMOUNT))
              OVER (PARTITION BY G.REGION_NAME, C.CATEGORY_NAME
                    ORDER BY T.YEAR_NAME))*100.0 /
             NULLIF(LAG(SUM(F.SALES_AMOUNT))
              OVER (PARTITION BY G.REGION_NAME, C.CATEGORY_NAME
                    ORDER BY T.YEAR_NAME),0),2)                     AS YOY_GROWTH_PCT,
       ROUND(SUM(F.SALES_AMOUNT)/NULLIF(SUM(F.UNITS_SOLD),0),2)    AS AVG_SELLING_PRICE
FROM   SALES_FACT F
JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME;
```

Power BI Visual: Q&A Visual (Growth & Share Dataset)

Q17: Q&A Semantic Dataset - Growth & Share (2024 Sample)

Region	Category	Sales	Units	ASP	Market Share %
North	Electronics	PKR 5.6M	22,293	PKR 250	7.47%
North	Clothing	PKR 4.4M	70,768	PKR 62	5.88%
North	Food & Beverages	PKR 4.0M	160,085	PKR 25	5.41%
North	Furniture	PKR 3.3M	9,772	PKR 334	4.38%
North	Sports	PKR 3.2M	31,880	PKR 99	4.23%
South	Electronics	PKR 3.1M	12,711	PKR 243	4.14%
Central	Electronics	PKR 2.8M	11,172	PKR 253	3.80%
North	Toys	PKR 2.7M	55,400	PKR 49	3.68%

Q18 – Q&A Visual (Comparative Trend)

Task: Comparative Trend Dataset

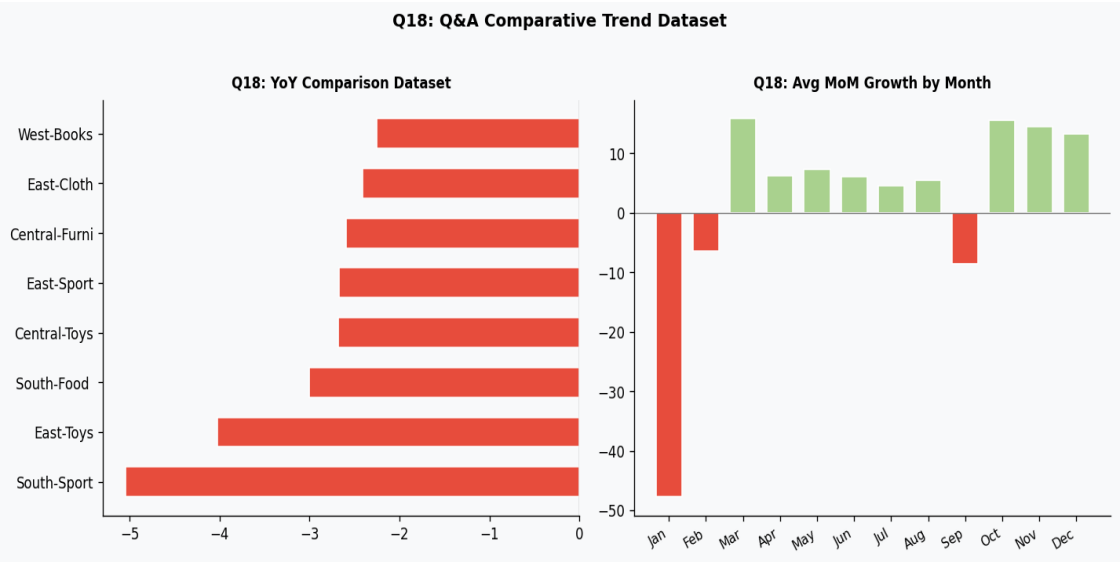
Comparative trend dataset with both month-over-month and year-over-year growth calculations to support natural language questioning.

SQL Query

```
WITH Monthly AS (
  SELECT G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.SALES_AMOUNT) AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   CATEGORY_DIM C ON F.CAT_KEY=C.CAT_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.REGION_NAME, C.CATEGORY_NAME, T.YEAR_NAME, T.MONTH_NUMBER
)
SELECT REGION_NAME, CATEGORY_NAME, YEAR_NAME, MONTH_NUMBER,
       MONTHLY_SALES,
       ROUND((MONTHLY_SALES - LAG(MONTHLY_SALES)
              OVER (PARTITION BY REGION_NAME, CATEGORY_NAME
                    ORDER BY YEAR_NAME, MONTH_NUMBER))*100.0 /
             NULLIF(LAG(MONTHLY_SALES)
              OVER (PARTITION BY REGION_NAME, CATEGORY_NAME
                    ORDER BY YEAR_NAME, MONTH_NUMBER),0),2)        AS MOM_GROWTH,
       ROUND((MONTHLY_SALES - LAG(MONTHLY_SALES,12)
              OVER (PARTITION BY REGION_NAME, CATEGORY_NAME
                    ORDER BY YEAR_NAME, MONTH_NUMBER))*100.0 /
             NULLIF(LAG(MONTHLY_SALES,12)
              OVER (PARTITION BY REGION_NAME, CATEGORY_NAME
                    ORDER BY YEAR_NAME, MONTH_NUMBER),0),2)        AS YOY_GROWTH
```

```
FROM   Monthly;
```

Power BI Visual: Q&A Visual (Comparative Trend)



Q19 – KPI (Rolling Performance Momentum)

Task: Rolling Performance Momentum

KPI dataset with rolling 6-month average sales compared against current month sales to measure performance momentum.

SQL Query

```
WITH MonthlyTotal AS (
  SELECT T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.SALES_AMOUNT)                                    AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY T.YEAR_NAME, T.MONTH_NUMBER
)
SELECT YEAR_NAME, MONTH_NUMBER, MONTHLY_SALES,
       ROUND(AVG(MONTHLY_SALES)
             OVER (ORDER BY YEAR_NAME, MONTH_NUMBER
                   ROWS BETWEEN 5 PRECEDING AND CURRENT ROW),2)   AS ROLLING_6M_AVG,
       ROUND((MONTHLY_SALES - AVG(MONTHLY_SALES)
              OVER (ORDER BY YEAR_NAME, MONTH_NUMBER
                   ROWS BETWEEN 5 PRECEDING AND CURRENT ROW))*100.0 /
             NULLIF(AVG(MONTHLY_SALES)
              OVER (ORDER BY YEAR_NAME, MONTH_NUMBER
                   ROWS BETWEEN 5 PRECEDING AND CURRENT ROW),0),2) AS VS_ROLLING_AVG_PCT
FROM   MonthlyTotal
ORDER  BY YEAR_NAME, MONTH_NUMBER;
```

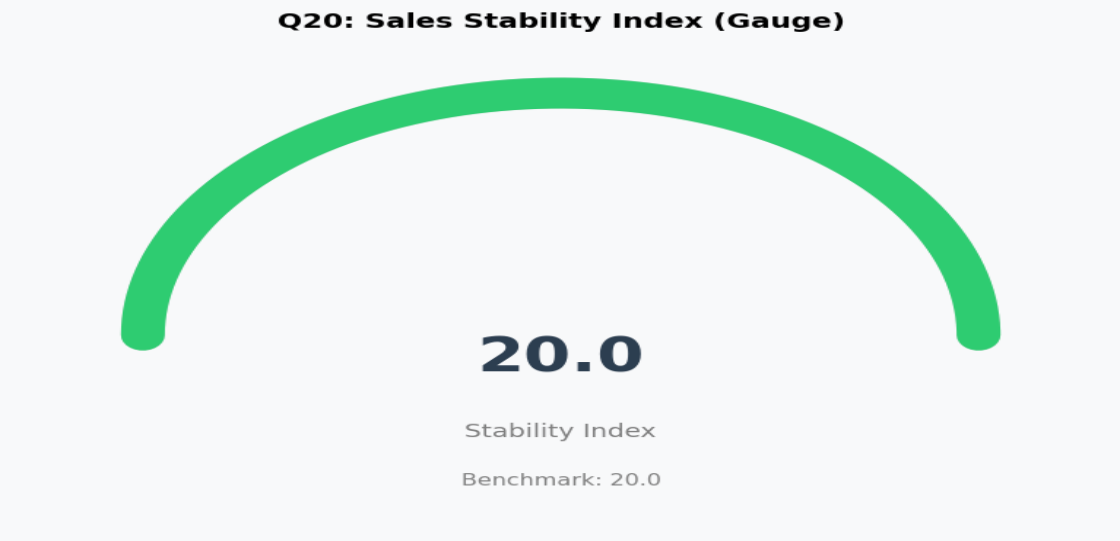
Power BI Visual: KPI (Rolling Performance Momentum)

Q20 – Gauge (Sales Stability Index)

Task: Sales Stability Index

Gauge dataset computing a Sales Stability Index (inverse volatility) per region, compared against the historical benchmark.

SQL Query

```
WITH MonthlyRegion AS (
  SELECT G.REGION_NAME, T.YEAR_NAME, T.MONTH_NUMBER,
         SUM(F.SALES_AMOUNT)                                        AS MONTHLY_SALES
  FROM   SALES_FACT F
  JOIN   GEOGRAPHY_DIM G ON F.GEO_KEY=G.GEO_KEY
  JOIN   TIME_DIM T ON F.TIME_KEY=T.TIME_KEY
  GROUP  BY G.REGION_NAME, T.YEAR_NAME, T.MONTH_NUMBER
),
Volatility AS (
  SELECT REGION_NAME,
         ROUND(STDDEV(MONTHLY_SALES),2)                             AS SALES_STD_DEV,
         ROUND(AVG(MONTHLY_SALES),2)                                AS AVG_MONTHLY_SALES,
         ROUND(1.0/NULLIF(STDDEV(MONTHLY_SALES),0)*100000,2)       AS STABILITY_INDEX
  FROM   MonthlyRegion
  GROUP  BY REGION_NAME
)
SELECT REGION_NAME, SALES_STD_DEV, AVG_MONTHLY_SALES, STABILITY_INDEX,
       ROUND(AVG(STABILITY_INDEX) OVER(),2)                         AS BENCHMARK_STABILITY
FROM   Volatility;
```

Power BI Visual: Gauge (Sales Stability Index)